

KDM5A fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes KDM5A gene fusions from the msk_impact_50k_2026 study, covering 10 fusion events. 3/10 fusions (30.0%) were in-frame. 5/10 fusions (50.0%) retained the PF08429 domain. Domain retention was tested with Fisher's exact test ($p=0.119048$, not statistically significant at $\alpha=0.05$). No literature benchmark is configured for KDM5A.

1 Results summary

1.1 Fusion partner frequency

Fusion-partner frequency was tabulated across 10 analyzed fusion events, identifying 9 distinct fusion partner genes. The most frequent partner was NINJ2, observed in 2 events.

1.2 Domain retention

PF08429 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 3/3 (100.0%) of in-frame fusions retained the domain, $p=0.119048$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p -value of 0.237624.

1.3 Domain disruption

Domain-disruption analysis was skipped: KDM5A has no `disruption_required_domains` configured; domain-disruption analysis was skipped.

1.4 Cutpoint detection

Cutpoint detection scanned 9 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 1311 aa (permutation-corrected $p=0.267327$, not statistically significant at $\alpha=0.05$). This is 94 aa from the nearest configured domain boundary at 1217 aa.

1.5 Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: KDM5A has no `group_field` configured; confidence-stats analysis was skipped.

1.6 Composite evidence score

Composite evidence score reported: `domain_disruption` sub-score excluded: the supplied result had no applicable p -value (e.g. `disruption_required_domains` not configured for this gene).

1.7 Exon retention

Exon retention reported: No target exon was supplied in `params` or `GeneConfig`.

1.8 Joint partner

Joint partner reported: KDM5A has no `gene_pair` configured; joint-partner analysis was skipped.

2 Tables

2.1 composite_score composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
COLGALT1	1	0.1	0.06241101320710366		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27943250471682274	1
DIPK1A	1	0.1	0.06241101320710366		0.9581105169340464		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27045618691697554	2
DHX37	1	0.1	0.06241101320710366		0.8440285204991087		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2460100448237746	3
NINJ2	2	0.2	0.06241101320710366		0.6154188948306596		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.23987941075196406	4
BMAL2	1	0.1	0.06241101320710366		0.4233511586452763		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.15586489585509622	5
USHBP1	1	0.1	0.06241101320710366				['domain_retention', 'recurrence']	0.08291409691231984	6
CACNA1C	1	0.1	0.06241101320710366		0.03208556149732622		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.0720222678948212	7
STAT5A	1	0.1	0.06241101320710366		0.031194295900178304		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.07183128240971809	8
OSBPL8	1	0.1	0.06241101320710366		0.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.06514679043110845	9

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2.2 cutpoint_detection cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
189	0	5	1	3	0.0	0.4444444444444445	0.3521825181113625
224	0	5	2	2	0.0	0.16666666666666669	0.7781512503836436
225	1	4	2	2	0.25	0.523809523809524	0.2808266095756941
664	2	3	2	2	0.6666666666666666	1.0	-0.0
759	2	3	3	1	0.2222222222222222	0.5238095238095237	0.2808266095756943
1311	2	3	4	0	0.0	0.16666666666666663	0.7781512503836437
1358	3	2	4	0	0.0	0.4444444444444444	0.35218251811136253
1486	4	1	4	0	0.0	1.0	-0.0

2.3 domain_retention frame_domain_contingency_table

3	2
0	4

2.4 domain_retention domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
auto_key_domain	PLU-1-like protein	9	5	1	3	0.015015015015015015

2.5 frequency Partner_gene_counts

Partner_gene	Event_count
BMAL2	1
CACNA1C	1
COLGALT1	1
DHX37	1
DIPK1A	1
NINJ2	2
OSBPL8	1
STAT5A	1
USHBP1	1

2.6 Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	is_protein_coding	is_transcript_coding	domain_retention	domain_retention_descriptives	disrupted_domain	disrupted_domain_descriptives	disrupted_domain_effect_on_fusion
EVT-P-0021409-T01-IM6-1	P-0021409-T01-IM6		COLGALT1-KDM5A	COLGALT1	unknown	three_prime	416253	24	1311	False	lost	0.0	False	PLU-1-like protein	COLGALT1 (NM_024656) - KDM5A (NM_001042603) rearrangement: t(19;12)(p13.11;p13.33)	chr19:g.1766921
EVT-P-0030719-T01-IM6-4	P-0030719-T01-IM6		KDM5A-BMAL2	BMAL2	unknown	three_prime	432924	15	664	False	retained	1.0	False	PLU-1-like protein	ARNTL2 (NM_001248002) - KDM5A (NM_001042603) rearrangement: c.285-4357:ARNTL2_c.1992:KDM5A	KDM5A
EVT-P-0030473-T01-IM6-5	P-0030473-T01-IM6		KDM5A-CACNA1C	CACNA1C	in-frame	three_prime	465876	6	225	True	retained	1.0	False	PLU-1-like protein	CACNA1C (NM_001129827) - KDM5A (NM_001042603) rearrangement: c.1114-2453:CACNA1C_c.673-173:KDM5A inv	

3 Figures

KDM5A domain-retention track

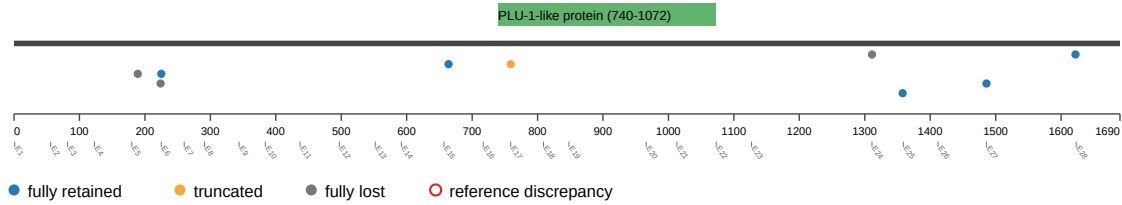


Figure 1: domain retention outliers

KDM5A fusion-transcript schematic

partner block → breakpoint → retained KDM5A portion (domain-colored, exon ticks)

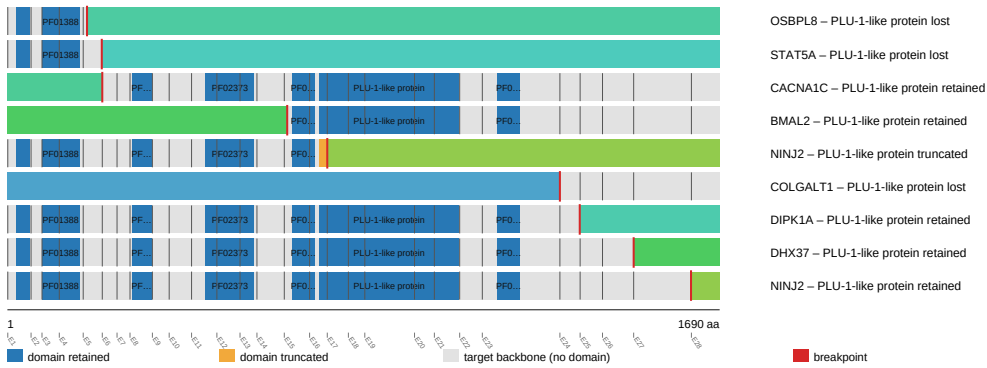


Figure 2: fusion schematic

KDM5A intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block



Figure 3: intragenic deletion schematic

Reference vs msk_impact_50k_2026



Figure 4: reference comparison